



Review

Evaluating current trends in stress and depression detection using artificial intelligence and machine learning



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ABSTRACT

Stress and depression are two major mental health-related concerns that affect people of varying ages, ranging from teenagers to working class people to senior citizens. The detection of these conditions helps the patients recover effectively which improves their quality of life. As behavioral, physiological, and digital trace data continues to be more accessible, computational models (especially AI-based ones) have become considered a promising solution to stress and mental health assessment. The current paper is a review of the recent research on AI-based stress and depression detection and tries to describe the way the field is developing. The diversity of the works is categorized broadly based on model type, input modality, data source, and evaluation strategy. Classical applications for machine learning remain quite prevalent, particularly for structured data such as EEG signals and questionnaire measurements. DL models and transformer architectures use of transformer-based architectures seems to be effective at physiological signal analysis and text inputs, but reported performance fluctuates. Vision-based models, particularly those using Convolutional Neural Networks (CNNs) on static image data such as facial scans or psychological drawings, are widely used. The use of Large Language Models (LLMs) and multimodal systems remains limited. Performance across reviewed studies was primarily evaluated using Accuracy, F1-score, and Root Mean Square Error (RMSE). Reported accuracies generally ranged from 74 % to 98 %, though results vary significantly depending on whether subject-independent validation was used. Some of the limitations in the literature we synthesized are - limited interpretability, inconsistent data standards, virtually no longitudinal modeling, and very little manifestation or clinical validation. This article unites these findings and identifies common gaps and recommends where the field might want to go next - including more standardized datasets, increased interpretability, and practical validation.

1. Introduction

Mental health, as defined by the World Health Organization, is a state of well-being in which each individual fulfills his or her own potential, can cope with everyday stressors, can work successfully and fruitfully, and is able to make a contribution to her or his community. It has been well established that one of the most important aspects of a person's health and well-being is their mental state¹.

There is increased evidence of the extent to which these issues have

become common. As an example, a quarter of students with diagnosed conditions note that they have serious mental health problems, and almost half of high school students complain of serious stress². Chronic stress is not simply an emotional burden, but it also affects physical health, leading to headaches, high blood pressure, irritability, sleeping disturbances and, in the long run, structural changes in the neural systems. Stress can also lead to a reduction of the size of the hippocampus, which can change the memory and emotional regulation, even when it is prolonged³.

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Stress and depression are untreated and spread to workplaces, families, and whole communities⁴. Productivity declines, relationships worsen and the healthcare systems are burdened. The inability to identify depression early is one of the greatest factors causing the increase in suicide rates all over the world^{5,6}. All these shocks add up, and the social and economic impact of these shocks is becoming more and more difficult to deny⁷.

Clinical diagnosis continues to depend on standardized self-report measures- some of the most popular are the HAM-D, the HADS and the DASS scales. The tools are helpful, and they are typically accompanied by the usual limitations: the responses can be biased, the stigma can lead to the inability to be honest, and the assessments are usually conducted relatively rarely⁸.

To better understand the contextual triggers of mental stress, several psychological models have been proposed. One widely referenced framework is the Job Demand-Control (JDC) model, later extended to the Demand-Control-Support (DCS) model. Developed by Karasek and Theorell, this model suggests that stress arises when high job demands coincide with low decision-making autonomy and insufficient social support. These models offer insight into how occupational environments can worsen mental health conditions⁹.

Other possibilities have also been made available by the developments in sensing and digital technologies. EEG enables the researchers to monitor the neural activity in real time, whereas less traditional sources, including patterns on social media or geospatial data, are being employed to analyze how the issues of mental health are evolving within the communities¹⁰⁻¹².

In this review, we examine the rapidly evolving landscape of computational approaches to mental health detection and diagnosis. Specifically, we explore:

- Machine learning methods using structured physiological or behavioral data,
- Deep learning techniques, including CNNs and RNNs,
- Vision-based models leveraging imaging data,
- Natural language processing and large language models (LLMs),
- Multimodal and hybrid systems that integrate diverse data types.

Most of the papers are devoted to one modality or technique, which does not provide a clear picture of the larger view.¹³ This review tries to cover that gap by bringing these individual threads together and providing a wider perspective of the current advances in machine learning, deep learning, natural language-based models, vision systems and the various hybrid combinations.

The rest of the paper is organized as follows: [Section 2](#) discusses the methodology, [Section 3](#) examines traditional machine learning strategies, [Section 4](#) discusses deep learning techniques, [Section 5](#) discusses computer vision methods, [Section 6](#) explores NLP, [Section 7](#) discusses multimodal and hybrid methods, [Section 8](#) critically synthesizes the findings, [Section 9](#) highlights future research directions, and [Section 10](#) concludes the study.

2. Methodology

The search was carried out to look for relevant research work, using leading research databases like Google Scholar, IEEE Xplore, SpringerLink, ScienceDirect, PubMed, PsycINFO, and arXiv. The search included English-language articles published from January 2017 to March 2025. The search terms used included those that relate to mental stress, along with search terms that denote computational techniques and data types. The search approach included a combination of search terms relevant to mental health, AI, and data types. The final search term was:

(Stress OR Depression OR Mental Health OR Anxiety) AND (Machine Learning OR Deep Learning OR Artificial Intelligence OR Neural Networks OR Transformer) AND (EEG OR Facial Expression OR Text OR

Multimodal OR Physiological)

Criteria for the inclusion of studies were defined as follows: those that analyzed psychological stress or related mental health indicators directly using computational methods were included. This would cover analyses that used classical or deep ML approaches for physiological signal or image-driven emotion recognition analyses, text classification for sentiment analytics based on transformer architectures, and multimodal approaches that combined multiple sources of information. Only English studies were included.

Our exclusion criteria were based on the following considerations: Purely conceptual papers, literature reviews, and other studies that were unrelated to psychological stress, for example, papers discussing mechanical stress, were excluded from further consideration. Duplicate entries and incomplete studies were also removed at this stage.

The initial search returned a total of 1781 studies. After duplicate removal and title-abstract level screening for relevance, 254 papers remained. These papers were subjected to a full-text review. Following this detailed evaluation, a final set of 78 studies was selected for inclusion in the review. Each of these studies met all predefined inclusion criteria and provided an adequate level of methodological and empirical detail ([Fig. 1](#)).

For methodological quality assurance, a systematic QA of potential articles has been performed. The articles had to be relevant, and a custom checklist has been used to assess studies according to their technical relevance, clarity of reporting, and robustness of validation. For more information on the specially designed checklist, refer to [Table 1](#). Studies not meeting the “critical” criteria were excluded.

3. Machine learning models stress and depression detection

Feature abstraction in psychiatric studies is generally plagued with the issue of the small data paradox. In their current application, deep learning models such as Convolutional Neural Networks and Transformers require the optimization of millions of parameters in an attempt to capture abstract features from the target data. Ironically, this convergence on useful features is contingent on the availability of big data, which rarely exists in the psychiatric community, as datasets rarely include more than 100 patients ([Fig. 2](#)).

There have been some recent works that support the use of support vector machines (SVMs) because of the generalization abilities presented in high-dimensional spaces. SVMs were shown to successfully distinguish between the groups of individuals with long-term stress and controls based on the alpha asymmetry index¹⁴. For instance, SVMs were used by Mutalib et al.¹⁵ on DASS-21 dataset scores obtained from over 600 university students, with the result illustrating that the prevalence of the condition was high; the model attained an accuracy level of 88.15 % with regard to distinguishing between the symptomatic and the asymptomatic individuals. SVMs were also used by Zulfiker et al.¹⁶ with favorable results obtained upon comparing the performance of the classifier with that of other classifiers on over 600 individuals with consideration of the psycho-social aspects. It can be observed that SVMs perform well where validation of the classification outcome can be attained.

Under the low-data setting, traditional machine learning techniques outperform modern deep learning architectures. Classification performance comparison experiments conducted utilizing the WESAD (Wearable Stress and Affect Detection) dataset and other physiological datasets illustrate that SVMs and RF always provide higher values of classification accuracy as compared to deep learning architectures when the data is not extensive (i.e., when N is relatively low). For instance, SVMs employing RBF kernels provide superior handling of high-dimensional feature spaces characterized by low values in the feature representation that corresponds to the extracted physiological features (e.g., heart rate variability spectral power, electrodermal activity peaks) ([Fig. 3](#)).

Further, the Random Forest algorithm has proved to be immensely

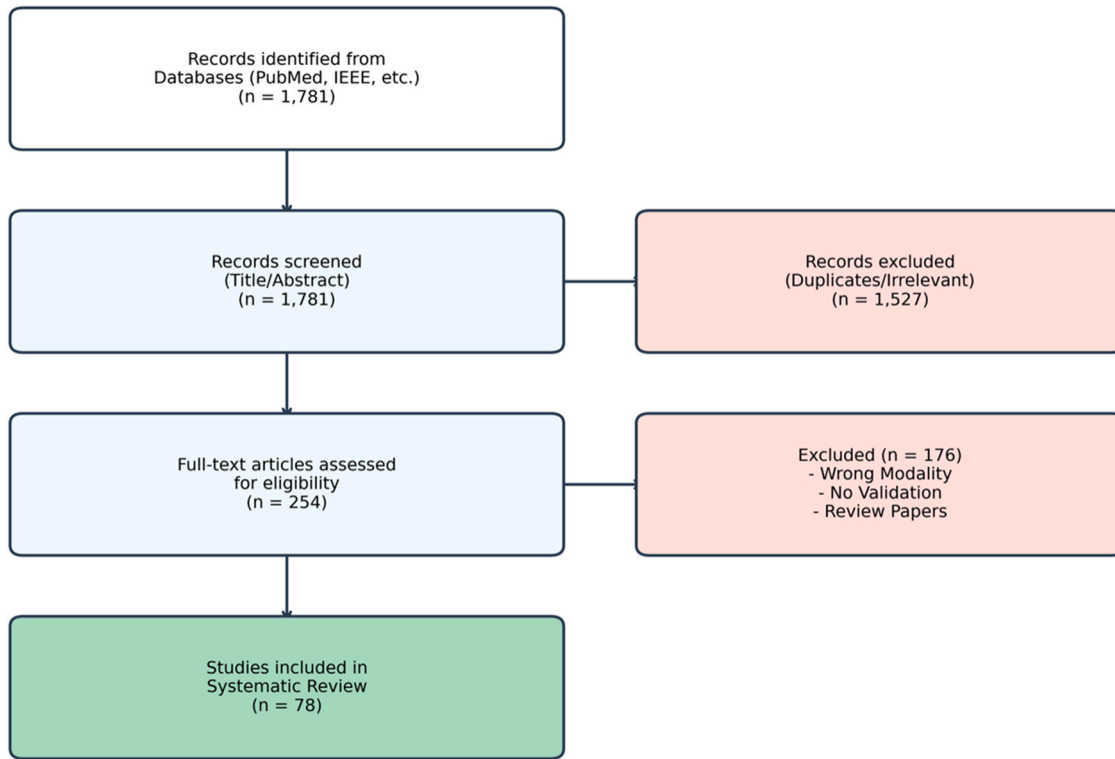


Fig. 1. PRISMA flow diagram of the study selection process.

Table 1

Quality assessment criteria (QC) checklist for study selection.

ID	Quality Criterion	Description & Requirement	Type
QC1	Relevance of Methodology	Does the study explicitly employ Machine Learning or Deep Learning for the detection of stress or depression?	Critical
QC2	Quantitative Rigor	Does the paper report standard classification or regression metrics to allow for valid comparison?	Critical
QC3	Data Transparency	Are the data sources, input modalities, and participant demographics clearly defined?	Critical
QC4	Validation Protocol (Risk of Bias)	Does the study clearly describe the validation strategy?	Supporting
QC5	Comparative Analysis	Does the study compare the proposed model against at least one baseline model or state-of-the-art benchmark to demonstrate technical improvement?	Supporting

practical within studies performed in 2024–2025. This is due to its ensemble learning strategy, wherein the randomness is effectively used to reduce individual decision tree variances. Priya et al.¹⁷ used k-nearest neighbors (kNN), on Google Form-based survey responses, with accuracy rates over 70 % in assigning levels of depression, anxiety, or stress. Purnamasari & Fernandya¹⁸ presented evidence that validated the effectiveness of using kNN on applications used by consumers, relying on EEG signals for stress detection in real-time. Furthermore, besides SVM, Zulfiker et al.¹⁶ analyzed kNN on the same data. They included specific results on balanced accuracy and specificity, suggesting that this algorithm is also practical or adequate in comparison with more complex alternatives, requiring less computation or having rapid application onboarding. These results illustrate that despite showing less competence compared with more refined alternatives, this algorithm still remains valid given its simplicity, fast performance, or easy implementation.

As is evident, classical models premised upon significant feature engineering, whereby knowledge about the domain is ingrained within the features themselves (e.g. computing the root mean square of successive differences, RMSSD, value for stress), will, when combined with lightweight classifier methods (e.g. SVM, k-NN), prove less computationally taxing and, thus importantly, more efficient than existing state-of-the-art DL systems, rather than necessarily less accurate. The aforementioned allow for the decomposition of signal processes that correlate to the activity of the autonomic nervous system (ANS), filtering out noise in the process.

On the other hand, studies based on end-to-end learning have shown high levels of accuracy, which have a tendency to reduce as the validation becomes more stringent. The use of raw signal values as the input to the deep learning system makes the system very prone to learning artifacts of the instrument, as opposed to the biological signal. The current trend in the year 2024 is the development of hybrid approaches, whereby one-dimensional CNN is used as the feature extractor for the traditional classifiers. One of the most important challenges associated with deep learning from a critical perspective in the domain is the instability of the validation. One of the shortcomings of many studies using deep learning is that they do not use a Leave-One-Subject-Out (LOSO) validation approach but instead randomly split the dataset (X: Y). There is high autocorrelation in physiological signals. The normal heart rate of the individual and the EEG alpha wave act as a unique biometric pattern. Therefore, when the dataset is randomly split, there might be overlap between the two sets. Deep learning models have high memory; therefore, they can easily learn the pattern of the individual instead of the pattern associated with the diagnosis. Recent comparisons have proven that when the LOSO approach is used for validation, the deep learning models perform much worse (reducing accuracy by 15–20 %), whereas the classical models (SVM/LDA) perform much worse but remain stable (Table 2).

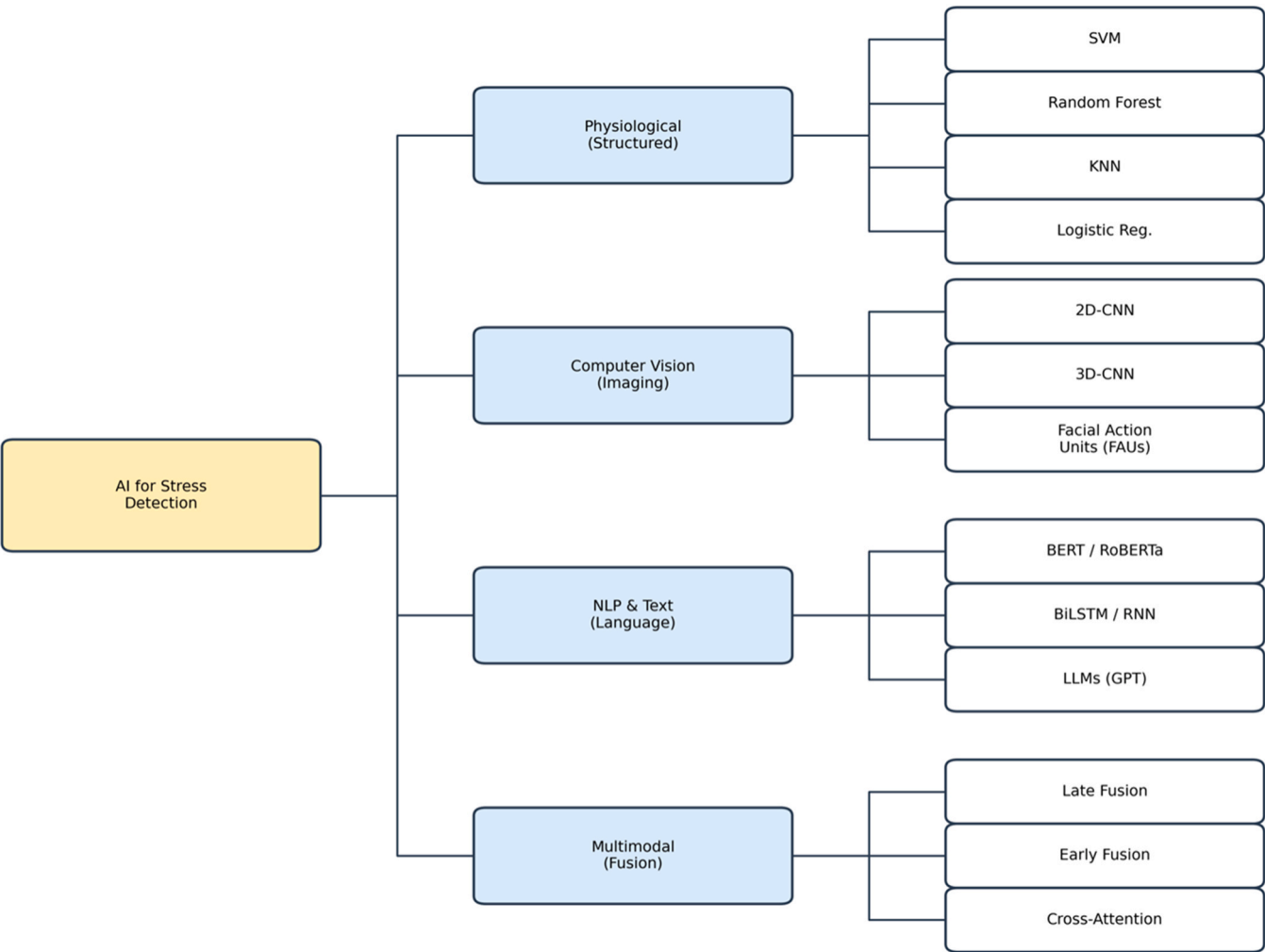


Fig. 2. Taxonomy of AI architectures for stress and depression detection.

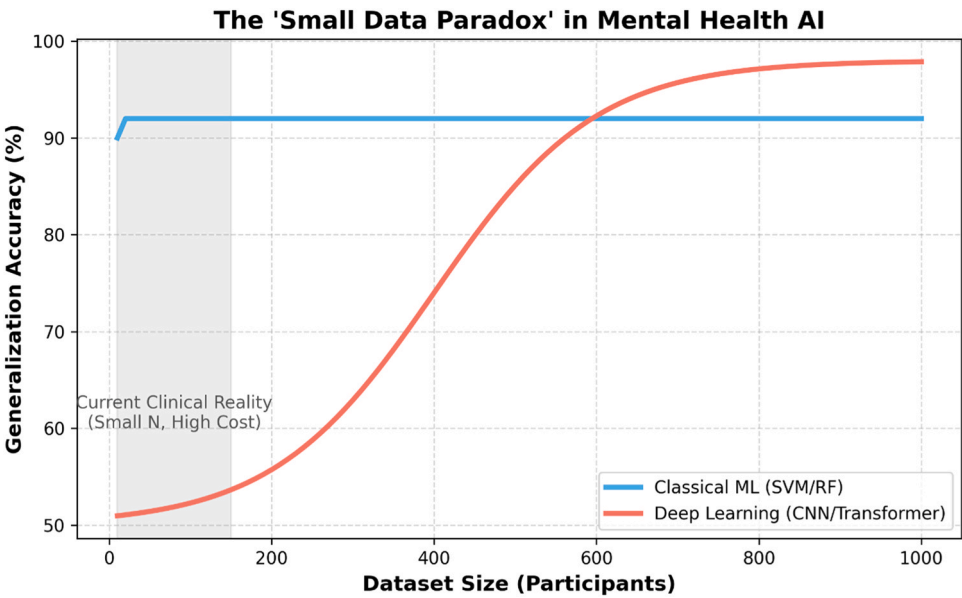


Fig. 3. The Small Data Paradox displayed while comparing average model performances used in this review work with respect to varying participant sizes.

Table 2
Summary of ML-based studies for stress and depression detection.

Dataset	Model Architecture	Training Approach	Performance Metrics	Key Results	Reference
PHQ-9 responses from 8 datasets including primary care and community samples	Logistic Regression (ML-based comparison of item pairs)	Supervised Learning on classification task (Depressive symptomatology)	AUC, Sensitivity, Specificity	Items 2 & 4 and 2 & 8 were found most predictive across datasets. Validated pairings outperform standard PHQ-2.	Glavin et al., ¹⁹
1094 individuals in Denmark with socio-demographic and mental health data	Bayesian Network	Hybrid model: data-driven + expert rule-based refinement	AUC	Hybrid model improved accuracy and correctly ranked disorder in top 2 choices for 90 %+ of patients.	Maekawa et al., ²⁰
Clinical + EEG data from treatment-resistant depression patients	Unspecified ML classifiers (pipeline to combine EEG + clinical features)	ML pipeline for classification and personalization	Planned: Classification Accuracy	Study protocol outlines the use of EEG-based ML personalization for pharmacological or stimulation therapies.	Pettorruso et al., ²¹
EEG-based stress data from brainwave sensors integrated with smartphone	k-Nearest Neighbor (k = 3)	Supervised classification	Accuracy (80 %)	Lightweight app-based detection system achieved reliable stress classification	Purnamasari, Fernandya ¹⁸
Depression dataset from Kaggle; standard features from socio-demographic survey	Random Forest, Logistic Regression, Naive Bayes	Standard supervised learning; evaluated with and without oversampling	Accuracy, Precision, Recall, F1-score	RF model performed best with 89 % accuracy and 90 % precision; oversampling improved performance	Hossain et al., ²²
HRV data from 37 MDD patients and 41 healthy controls, recorded during five 5-min experimental phases.	Support Vector Machine (SVM)	Supervised learning on 100 HRV features across all phases	Accuracy: 74.4 %, Sensitivity: 73 %, Specificity: 75.6 %	SVM model effectively differentiated MDD from healthy participants using HRV indices.	Byun et al., ²³
206 student participants; Perceived Stress Scale (PSS) questionnaire responses	Random Forest, SVM, Naive Bayes, k-Nearest Neighbors	Feature extraction followed by supervised classification using four ML algorithms	Accuracy: 85.71 %	RF and SVM provided the best performance across evaluated metrics.	Ahuja, Banga ²⁴
18 healthy males (20–24 years old) undergoing TSST protocol for stress elicitation	Three binary SVM classifiers + ECOC (Error-Correcting Output Codes)	Multiclass classification with MATLAB implementation	Accuracy: Avg 94.79 %, decreasing with increasing stress levels	System capable of classifying multiple stress levels with high accuracy	³⁾
EEG data from 13 depressed and 12 healthy individuals aged 20–50	SVM with Leave-One-Out Cross Validation	Supervised learning using extracted EEG features	Accuracy: 84 %, Sensitivity: 83.33 %, Specificity: 91.3 %	SVM-based system achieved high specificity and sensitivity using EEG signals	Mantri et al., ⁵
WESAD dataset with wearable data from RespiBAN and Empatica E4 (15 volunteers)	SVM Classifier	Supervised binary and three-class classification using statistical features	Accuracy: Up to 93.20 % (binary), 81.65 % (three-class)	SVM yielded high accuracy in distinguishing stress states from physiological data	Bobade, Vani ²⁵
HRV and ECG data collected from participants; seven HRV features extracted	SOM + SVM (two-stage classification)	Hierarchical classification using clustering and supervised SVM	Accuracy: 91 %	Two-stage SOM-SVM hybrid model effectively identified four stress categories	Vanitha et al., ²⁶
EEG data from 20 healthy and 15 depressed participants (ages 18–55)	SVM with Gaussian RBF kernel	Kernel Target Alignment (KTA) + SVM classification	Accuracy: 80 %	Optimized channel selection with KTA improved EEG-based classification accuracy	Shen et al., ²⁷
Physiological sensor data from IoT devices used to monitor student stress	Multinomial Naive Bayes	Real-time cloud and fog layer emotion classification	Accuracy: 90.11 %	NB model integrated in IoT framework achieved high accuracy in live scenarios	Singh et al., ²⁸
EEG data collected from 42 healthy participants under MIST paradigm	Naive Bayes Classifier	Feature selection via SFS and SBS, followed by classification	Accuracy: 94.6 %, Sensitivity: 98.3 %, Specificity: 93.3 %	NB model accurately detected both binary and multilevel stress conditions	Subhani et al., ²⁹
EEG signals from 15 healthy and 15 depressed individuals	KNN (using orthogonal wavelet filter bank features)	Wavelet sub-band feature extraction + KNN classification	Accuracy: Left 98.5 %, Right 98.7 %	KNN combined with TCOWFB showed near-perfect classification performance	Sharma et al., ³⁰
ECG data from 20 university students; six emotional states induced	KNN (k = 3, k = 4), evaluated with 5 and 10 frames	Framed ECG feature extraction + classification	Max Accuracy: 86.1 %	Ten-frame ECG signal segmentation improved classification of stress states	Zheng et al., ³¹
80 M tweets from the US (Jan–Apr 2020), PHQ-9 stress index	Naive Bayes with CorExQ9 symptom extraction	Social media mining and unsupervised label generation	Accuracy: 80.76 %	NB showed reasonable accuracy on weakly labeled Twitter data	Li et al., ¹⁰

4. Deep learning (DL) for stress and depression detection

There is growing interest in deep learning techniques for mental health detection that are moving in the direction of designing architectures for not just classifications but for understanding temporal and cross-modal interactions as well. One study proposed designing a temporal attention convolutional neural network for estimating levels of depression based on DAIC-WOZ data. In this way, the approach leverages temporal matching of different modalities, hence predicting those parts of interactions where depressive expression is of prime importance, like pauses, hesitation in speech, or flat affect. In this context, it is essential to note that they are more concerned about temporal explanation than prediction.³²

In a complementary approach, Wei Liu et al.³³ used BiLSTMs to

harness electroencephalography (EEG) signals. The system recognizes spatial correlations in channels, in addition to temporal dynamics, making it both biologically plausible and credible. The paper also points out the relevance of wearables in supplying real-time data when combined with deep sequence models.

Other techniques, such as attention-driven CNN-RNN fusion architectures for stress state classification and transformers with multimodal fusion capabilities, indicate a trend towards the growing use of sequence awareness in modeling the behavioral as well as the physiological signals in the concerned studies. It is as if the trend in the concerned areas is to move from simple classifiers and toward architectures similar to a clinician's observational assessments of human behavior over time, and sometimes using multiple modalities.

One of the criticisms constantly prevalent in the 2024–2025

literature is that simple Long Short-Term Memory (LSTM) networks are inefficient in modeling non-local dependencies critical to detecting conditions of chronic stress or depression. As shown in LSTMs where information is sequentially processed through a network from end to end, it is highly probable that an information bottleneck is formed where activities in previous events (like morning sleep inertia) do not affect downstream states (like evening cortisol levels). Conversely, as shown in experiments conducted by Hu et al.,³⁴ on F1-score validation (approximately 98 %) achieved in Transformer encoder variants, non-local dependencies are efficiently addressed with decreased dependencies in longer temporal states due to global self-attention mechanisms in place in these variants that are ignored in LSTMs.

Imbalanced mental health data sources tend to have a minority group consisting of people with mental health issues. The traditional method of applying the Synthetic Minority Over-sampling Technique (SMOTE) oversampling approach is increasingly coming into question for deep learning. Studies by both Aldayel & Al-Naffjan³⁵ and Daud et al.³⁶ have shown that for nominal accuracy, SMOTE can actually add linearity that the deep model memorizes rather than capturing meaningful information that boosts performance metric accuracy. Shen et al.,³⁷ as well as Zhu et al. in 2024, proved the effectiveness of Focal Loss over SMOTE in detecting depression by appropriately altering the loss function to give less importance to the easy negative classes, thereby increasing sensitivity without generating synthetic data. This is a milestone in the respective field, accepting the fact that generating more data synthetically might not be as effective as the optimal loss function in handling the existing prevalence of diseases in reality (Table 3).

5. Vision-based approaches

Vision-based methods have proved to be highly effective in estimating nonverbal cues related to stress, fatigue, and depression.

Feng et al.⁴⁴ utilized GRU neural network architectures for processing sequences of facial landmarks derived from video recordings of interviews. In contrast to many CNN architectures, which process images frame by frame, independently, this strategy processes patterns of movements, which can help track muscular changes that might be indicative of moods disorders. The benefit of this strategy is that it utilizes lightweight data, namely landmarks.

Convolutional Neural Networks have a naturally existing bias toward locality and translation invariance, leading to the data-efficiency of the networks.⁴⁵ According to Gong et al.⁴⁶ and Sajjad et al.⁴⁷, in the relatively small-scale depression datasets (such as the AVEC and DAIC-WOZ challenges), CNNs have always beaten the Vision Transformers (ViTs). Since ViTs fix an image as a series of patches, they lack these existing biases and, as a result, have to learn the inter-region relationships after heavy pre-training on a large-scale dataset (e.g., the JFT-300M dataset), as highlighted by Xue et al.⁴⁸, before the ViTs can converge on depression-related micro-expressions, fortunately still the stronger architecture in the existing depression groups.^{49,50}

Validated accuracy in many studies clouds large discrepancies in relation to race and gender differences.⁵¹ On these grounds, face expression recognition FER tasks were rigorously audited by Chhwa et al.⁵², wherein FER models were recognized to wrongly categorize the expression of Black men in their neutral form as negative (angry/-contempt), which is not just an implementation problem but is a health/safety risk as well because it can cause an excessive diagnosis of the condition in that group of people.

Moreover, the utilization of static frame analysis is a simplifying assumption. Jiang et al.⁵³ and Pan et al.⁵⁴ pointed out that 2D-CNN methods used in depression recognition are inappropriate because the appearance of psychomotor retardation is characterized by the dynamic expression of depression (e.g., the duration of smiling), not static expressions. The idea behind 3D CNNs and combined CNN-RNN models partially resolves this problem⁵⁵. However, they introduce more complexity when it comes to the demand for more extensive amounts of data. Thus, the community has reached the level of a trade-off between the trainability and complexity of 2D and 3D/ViT models. Recently, the work of Kim et al. (2023) and Roy et al.⁵⁶ combines the use of gaze, thermal, or facial features with tree-based or 3D-CNN models that are understandable. This demonstrates an increasing effort to bridge the existing gap between affective computing and the interpretability of results, where visual signs are no longer mere features but are linked to concrete psychophysiological patterns. The area gradually moves towards the use of hybrid input and lightweight deployment methods (Table 4).

Table 3
Summary of DL-based studies for stress and depression detection.

Dataset	Model Architecture	Training Approach	Performance Metrics	Key Results	Reference
EEG signal data from clinical depression studies	Graph Neural Network (spatial topology) + Temporal Graph Network	Supervised classification using depression severity labels	Accuracy, F1-score	Graph-based architecture improved detection performance by modeling cross-channel dependencies in EEG.	Liu et al., ³⁸
Cognitive EEG task data from healthy and depressed subjects (internal dataset)	CNN-LSTM based fusion of resting and task EEG	Deep supervised learning for binary classification	Accuracy: 92.1 %, AUC: 0.91	Fusion of resting and task EEG improves generalizability and performance in MDD classification.	Peng et al., ³⁹
Audio speech signals from depression interviews (Bangkok dataset)	CNN + Transformer classifier	Supervised voice modeling	Accuracy: 84.3 %	Model achieved strong performance on audio-only input; potential for non-invasive, speech-based depression tools.	Suparatpinyo and Soonthornphisaj ⁴⁰
Review study summarizing multiple datasets: AVEC, RECOLA, and others	CNN, 3D-CNN, LSTM (from evaluated models)	Trained models for regression or binary depression classification	Not directly reported (review)	Temporal models with multimodal fusion perform best for affect recognition in depression.	Liu et al., ³³
Public survey data from Black youth with depression scores	LSTM, Random Forest	Multimodal learning combining structured text + features	Accuracy, AUC (not numerically reported)	Deep models outperformed classical ML in diverse demographic groups.	Zhong and Yadav ⁴¹
Multimodal physiological dataset (ECG, GSR, EMG) from induced stress/depression study	Multitask deep neural network	Multitask supervised learning	Accuracy: 87 % (stress), 84 % (depression)	Joint learning of stress and depression tasks improved generalization across overlapping physiological features.	Ilias et al., ⁴²
HRV features collected via wearable sensors from mental health study participants	BiLSTM	Supervised classification (mental health disorder detection)	Accuracy: 89.2 %, F1-score: 0.88	BiLSTM captured temporal dynamics of HRV signals and outperformed traditional models.	Singh and Kumar ⁴³

Table 4
Summary of vision-based studies for stress and depression detection.

Dataset	Model Architecture	Training Approach	Performance Metrics	Key Results	Reference
DAIC-WOZ depression video interviews (audio-visual corpus)	MDN (CNN + attention modules)	Supervised learning on PHQ-8 regression task	RMSE, Pearson Correlation Coefficient	Temporal dynamics modeled via CNN + max-diff network improved PHQ-8 score estimation.	De Melo et al., ⁵⁷
Annotated psychiatric interview video transcripts	Emotion recognition model with multi-label classification	Supervised sequence modeling	F1-score (emotion detection)	Emotional content over segments of the interview helped detect depressive tendencies.	Oh et al., ⁵⁸
Facial expression video frames from clinical interview corpus	GRU with attention on landmark sequences	Sequence classification	Accuracy, F1-score	Subtle facial motion patterns provided high predictive accuracy for depressive state.	Feng et al., ⁴⁴
Selfie images with metadata and DASS-21 labels	Vision-language foundation model	Zero-shot + fine-tuning	AUC (no fixed value reported)	Model enabled remote depression screening using selfies and simple metadata.	Basilio et al., ⁵⁹
Infrared facial thermograms from clinical population	Random Forest + interpretable feature selection	Supervised classification	Accuracy: 82 %	Thermal asymmetries and facial heat patterns were useful for depression classification.	Roy et al., ⁵⁶

6. NLP / LLM approaches

In the more recent studies of NLP in the context of mental health, the focus turned to language context, temporal dynamics, and style classification. Zirikly et al.⁶⁰ utilized the transformer architecture on the Timeline of Reddit to identify depression, stressing the importance of sequential thinking over point classification. The results indicate an overall trend in the community: the presence of mental health cues is identified via simpler systems, and models that identify posting patterns will ultimately provide more reliable results.

Orabi et al.⁶¹ analyzed Twitter data and included elements of punctuation, syntax, and emotion expression in XLNet. They show how, with domain-specific linguistic elements, even with a fairly informal and noisy medium, large-scale language models are capable of distinguishing content related to stress. They have demonstrated how transformers could be supplemented with simpler downstream classifiers to obtain more stable results.

Other research efforts, such as the studies by Varadarajan et al.⁶² and Lee et al.⁶³, continue to explore in the same direction by considering emotion lexicons and multi-label scenarios such as cognitive distortions. These studies in combination thus identify an interesting pattern: it appears that the efficacy of transformer-based embeddings can be greatly improved by applying domain-filtering techniques, hierarchical modeling of context, or psychological theory-based elements.

Current analyses performed by Umapathi et al.⁶⁴ confirm that large LLMs tend to rely on the generation of symptoms based on a given

diagnostic hypothesis (confirmation bias). That means that an LLM may include suicidal ideation information from a patient diagnosed with depression simply because there's a statistical correlation between the two terms throughout the LLM's dataset. Bai et al.,⁶⁵ measured the extent of the above problem by highlighting that LLMs designed with a medical domain tend to demonstrate a higher rate of hallucinations compared with general LLMs because of overfitting.

At the same time, the application of NLP in mental health faces the issue of the privacy-utility trade-off. The kind of text data necessary for diagnosing depressions, such as journals and therapies, is extremely personal. Nagy et al.⁶⁶ have explored the impact of federated learning and differential privacy methods, showing that the provision of very tight bounds affects diagnosis negatively. In particular, Karpontinis et al.⁶⁷ showed that the noise level in DP-CARE, introduced as defense for membership inference attacks, negatively affects the recall in such cases as suicidality assessment. This leads to a moral deadlock, as the preservation of the privacy of patients via high-noise DP ($\epsilon < 3$) hurts the ability of the model to spot the most severe, life-threatening instances (Table 5).

7. Multimodal / hybrid approaches

A multimodal platform supports the combination of different input signals such as facial video, audio, EEG recordings, and text into a unified predictive model. Wang et al.⁷⁰ introduced a transformer-based architecture with dynamically weighted cross-attention between

Table 5
Summary of NLP and LLM approaches for stress and depression detection.

Dataset	Model Architecture	Training Approach	Performance Metrics	Key Results	Reference
Reddit post timelines from mental health subreddits	Transformer (BERT) with temporal sequence modeling	Fine-tuned pretrained transformer on depressive vs. control timelines	Accuracy, F1-score	Temporal models captured subtle progression of depressive symptoms over time better than flat classifiers.	Zirikly et al., ⁶⁰
Twitter posts labeled with stress, anxiety, depression	XLNet + linguistic feature ensemble	Fine-tuned transformer + traditional classifiers	Accuracy: 85.3 %, F1-score: 83.1 %	Model handled short informal texts and boosted robustness using syntactic/emotional features.	Orabi et al., ⁶¹
Mental health therapy dialogue transcripts	Transformer + multilabel logistic regression	Supervised classification with cognitive label tagging	Micro-F1: 0.72	Enabled interpretable detection of psychological distortions for different therapy sessions.	Lee et al., ⁶³
PHQ-9 items from large multi-cohort study	Logistic Regression (lightweight NLP model)	Binary classification using item pairs	AUC, Sensitivity, Specificity	Pairs like item 2 & 4 proved more predictive than standard PHQ-2 screening tool.	Glavin et al., ¹⁹
Mental health tweets labeled for depressive expression	BERT + confidence calibration layer	Supervised transformer training with post-hoc calibration	ECE (Expected Calibration Error), Accuracy	Calibration improved trustworthiness of predictions on noisy, real-world social media inputs.	Ilias et al., ⁴²
Reddit posts across mental health forums	BERT + linguistic feature injection	Fine-tuned transformer with syntactic/lexical augmentations	F1-score: 0.84, Accuracy: 87 %	Combining pretrained models with feature engineering led to better generalization across subreddits.	Rizwan et al., ⁶⁸
Therapy and support forum transcripts (multi-label emotion tags)	GPT-style transformer with multi-label classification head	Zero-shot and fine-tuned settings	Micro-F1: 0.75	LLMs can perform on par with task-specific models in emotion recognition, even in noisy settings.	Tan et al., ⁶⁹

modalities using CMU-MOSEI as a training dataset in 2022. This is particularly effective when one of the input sources has noisy or uninformative information to offer to the model due to various environmental conditions in real scenarios (Fig. 4).

Khan et al.⁷¹ stacked layers of CNN and LSTM networks over physiological signals and used saliency maps to demonstrate the role of various regions of the inputs in stress classification. In addition to improving architecture, the ability to convey signal-level explanations increases the applicability of the AI system in real-world settings.

Later research by Majumder et al.⁷² and Huang et al.⁷³ used tensor fusion, modality gating, or graph neural networks to explore the relationships between the features over time and between the different modalities. Such designs are more advanced than the concatenation of features to explore the psychological unity between the modalities.

Fan et al.⁷⁴ argued that the existing Late Fusion method is flawed because it considers each modality an independent vote, making it difficult for the model to capture the inter-modal relation (for instance, a smiling face but an expressionless voice). For the case of missing video information, the model fails to utilize the existing audio information to predict the missing information.

To tackle this issue, there have been some recent advancements in the Generative Imputation and Retrieval-Augmented techniques. Fan et al.⁷⁴, and others proposed models named RAMER (Retrieval Augment for Missing Modality), in which the model retrieves and generates the missing feature vectors depending upon the existing modality, as opposed to setting missing data to zero. Additionally, graph neural networks have been used to adaptively re-weight existing nodes in case of a sensor failure problem. However, this approach brings along the problem of synthetic diagnosis. Through imputing the missing physiological or visual observations, the model makes a diagnosis on the basis of predicted rather than observed human behavior. In this manner, if a model diagnoses high stress associated with its predicted HRV signal, which was lost due to the disconnection of the sensor, it might end up in creating a self-fulfilling prophecy that aligns with its training distribution rather than addressing the patient's present state because technical robustness gains in tasks like F1 score are at the expense of clinical validity.(Table 6).

8. Results and discussion

From the synthesis of the papers, there are few interesting results among model performances. As seen in Fig. 5, there exists a tremendous level of fragmentation in the field, with the unimodal approaches still being the most widely adopted (65 % of studies surveyed), mainly centered on physiological and computer vision modalities. When evaluating the effectiveness of such paradigms, Fig. 6 illustrates a significant

performance difference, pointing to classical machine learning models such as SVMs and Naïve Bayes, with maximum attainable accuracy (~94.8 %) drastically outperforming more sophisticated Deep Learning models such as CNNs (~84.3 % and 3D-CNNs (~81.6 %). The existing difference is justified by the Small Data Paradox, described in Fig 3. While more sophisticated models like Deep Learning have a greater capacity for improvement, their large dataset requirement necessary for a smooth and successful training process, far surpasses the available dataset size in the prevailing mental health research environment with datasets belonging to the "Clinical Reality" set, making it easier for classical algorithms to adapt and generalize compared to neural models apt to significantly deteriorate on small data samples.

Traditional machine learning models such as support vector machines (SVM), random forests, k-nearest neighbors (KNN), and gradient boosting are still being used in newer studies, particularly in work based on structured or low-dimensional physiological data. These include self-reported questionnaire scores (e.g., DASS-21, PHQ-9), handcrafted features from EEG signals, and demographic variables. Their relative interpretability and ease of deployment make them especially appealing in low-resource settings and applications with smaller datasets. Feature selection methods such as recursive feature elimination and mutual information ranking are frequently used to improve signal-to-noise ratio and prevent overfitting. However, these models often rely on assumptions of feature independence and are less suited to capturing complex, non-linear relationships. Furthermore, generalization remains a concern in imbalanced or demographically skewed datasets.

Deep learning techniques have seen growing adoption in recent years. CNNs remain prevalent in visual analysis, while long short-term memory (LSTM) networks, gated recurrent units (GRUs), and 1D CNNs are frequently applied to time-series data such as EEG or audio recordings. Autoencoders and sparse encoders are used for dimensionality reduction and anomaly detection, particularly in unsupervised settings. Transformer-based architectures are beginning to appear in text-based applications, including social media mining and clinical note classification. However, most of these models operate in a unimodal fashion. Few studies attempt to integrate visual, physiological, and textual data into a unified diagnostic framework. This siloed approach limits the richness of inferences that could otherwise emerge from multimodal fusion.

Many studies use static image information, which could be magnetic resonance images (MRI), facial images, or psychological drawings like the House-Tree-Person Test. The models used are mostly convolutional neural networks (CNNs), which are preferred due to their feature extraction and pattern recognition capabilities. The models are generally used to identify depressed patients from non-depressed ones through structural/morphological differences. In certain cases, attention mechanisms and saliency maps are utilized to identify important areas of interest in the images. However, the major drawback of these models lies in their inability to deal with dynamics, as they are designed to process individual images. This makes them unsuitable for use in applications involving long periods of time, like monitoring symptom progressions and emotional expression.

NLP techniques, especially those based on large pretrained language models, have started to be used for analyzing the user content of websites like Reddit, Twitter, and internet-based therapy communities. Some common examples of NLP applications include sentiment analysis, emotion labeling, and suicidal risk prediction. Many research studies have made use of the transformer-based network, which includes the application of the BERT and RoBERTa algorithms. Some research studies have made use of ensemble pipelines, which in turn rely upon semantic, syntactical, as well as lexicon-based features. Although significant advancements have occurred in the application of NLP, the application areas of NLP itself have remained quite narrow, which may include only a single-step classification without context. There is a need to conduct research in the application of multi-turn dialogue, patient information encoding, as well as time-evolving symptom encoding, which have real-

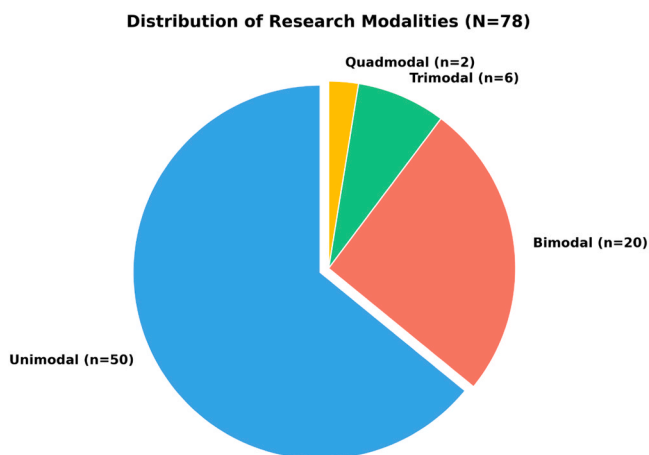


Fig. 4. Modality distribution of the research works reviewed in this paper by virtue number of model input.

Table 6
Summary of multimodal and hybrid approaches for stress and depression detection.

Dataset	Model Architecture	Training Approach	Performance Metrics	Key Results	Reference
CMU-MOSEI (video, audio, and text of emotional speech)	Multimodal Transformer with attention-based fusion	Supervised learning with joint modality training	F1-score, Accuracy	Cross-modal attention improved performance over independent fusion; robust to missing modalities.	Wang et al., ⁷⁰
Custom dataset combining EEG and facial video features	CNN-LSTM + saliency module	Early fusion of spatial (face) and temporal (EEG) inputs	Accuracy, Saliency Visualization	Fusion model interpretable and outperformed single-modality baselines for stress detection.	Khan et al., ⁷¹
DAIC-WOZ interviews with audio + text + facial expression	Hierarchical attention-based multimodal network	Supervised training using PHQ-score ground truth	F1-score, RMSE	Gated attention improved feature selection across modalities; better score prediction.	Majumder et al., ⁷²
Multimodal patient records (text, clinical notes, labs)	Graph Neural Network for symptom-interaction modeling	Supervised symptom classification	AUC, Macro F1	GNN architecture captured interactions between diverse symptom types for improved prediction.	Huang et al., ⁷³
Survey + text + speech multimodal inputs from adolescents	EmbraceNet + auxiliary attention layers	Multi-task supervised training	Accuracy, Multimodal Consistency	Framework generalized well across missing modalities and task types.	Du et al., ⁷⁵
DAIC-WOZ: audio recordings + transcripts	Dual encoder LSTM with attention fusion	Supervised regression (PHQ-8 score prediction)	MAE: 3.15, RMSE: 3.79	Speech and transcript fusion improved depression score prediction over individual inputs.	Zhao et al., ⁷⁶
Audio, facial expressions, and questionnaire responses (custom corpus)	Multimodal Reasoning Network (MMRN)	Supervised learning for binary diagnosis	Accuracy: 90.2 %	Cross-modal attention reasoning mimicked clinician-like behavior and improved interpretability.	Huang et al., ⁷⁷

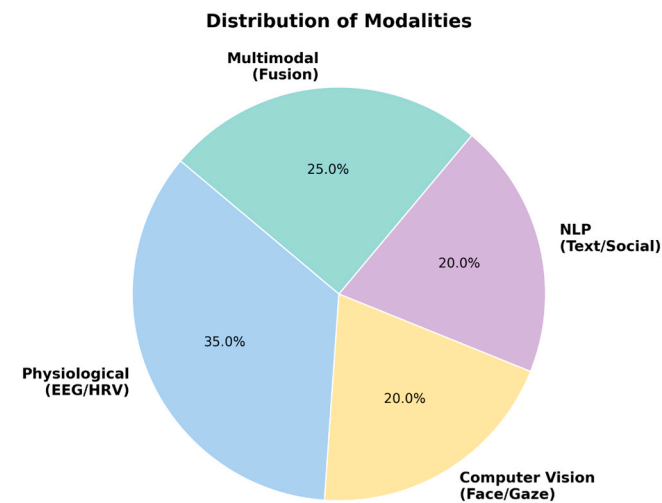


Fig. 5. Modality distribution of the research works reviewed in this paper by virtue of type of model input.

world significance.

Explainability is an area that is not adequately explored in this field despite the fact that the model results are not clinically acceptable if they are not interpretable. Though there are various studies that attempt to introduce interpretability, the application of SHAP, LIME, or attention maps is very limited. In fact, these tools are very often tacked on as an afterthought as opposed to being an inherent aspect of the design. Not only that, but these explanations are rarely checked either for adherence to psychiatric diagnosis or for validity with mental health professionals. Without such a link, there is limited utility, reducing clinical acceptability (Figs. 7 and 8).

Multimodal learning has been rarely explored given the potential benefits. The few studies attempting multimodal fusion are invariably late fusion-based, combining the independent model outputs for different modalities. Late fusion is easy to do; however, it fails to model the relationship across modalities and their synchronization. The use of sophisticated methods such as early feature fusion or intermediate feature fusion through attention or embeddings has been very uncommon. Moreover, no standard protocols for multimodal learning are defined for comparisons across studies.

Methods using physiological signals like electroencephalography (EEG) or electrocardiography (ECG) have been found to perform well in experimental environments.⁷⁸ The common signal processing tasks include filtering, time and frequency analysis (e.g., wavelet or fast fourier transforms), and statistical extraction, followed by machine learning (ML) and/or deep learning (DL) classification. These models have the problem of reproducibility despite the reported high classification rates. The used data is generally limited in size, and the demographics used during modeling remain remarkably narrow. Further, the disparities in the configurations of electrodes, processing, and environments of acquisition make cross-validation impossible. Additionally, the hardware, such as the use of multi-electrode caps or medical-grade equipment, limits practical application.

Many of the models have application in diagnostic classification, usually in terms of binary assignments (e.g., depressed vs. non-depressed subjects). Although this is easier to assess, it does not capture the reality of mental health issues, which are complex in the sense of intensity as well as the potential presence of multiple conditions. Very few studies have explored the application of regression-based scoring for intensity, multi-class, or multi-label classification.

Challenges primarily involve label quality and the reliability of the datasets. Most studies rely on supervision methods like the use of hashtags as labels, crowd-sourced labels, and output of zero-shot machine models. These not only make it easy to generate datasets at scale but pose challenges in the form of noise and bias in the labels. There is little to no measure of inter-rater reliability or evaluation by experts after annotation tasks in most datasets. Additionally, these datasets contain little to no linguistic diversity in terms of regions and other demographic factors.

Finally, issues that are related to deployment are barely discussed in literature. Very few reports are found regarding performance when constrained by real-time latency, power use, or personalization. Issues related to mobile deployment, compatibility with edge computing, and preserving privacy by means of federated learning are very seldom discussed. It should be noted that studies related to integration into clinical practices, clinician-in-the-loop systems, or feedback into the model are absent from the literature.

9. Future directions

The current state of the field reveals an ever-growing need for spatiotemporal models when forecasting mental health states, especially

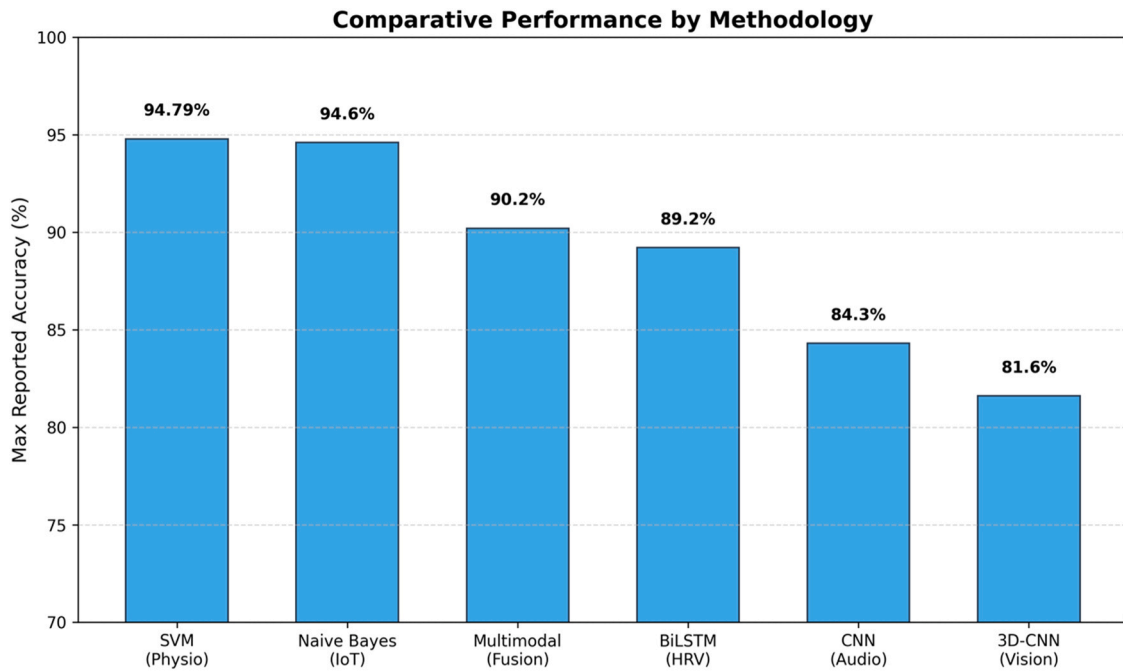


Fig. 6. Comparative analysis of maximum reported accuracy across different methodologies.

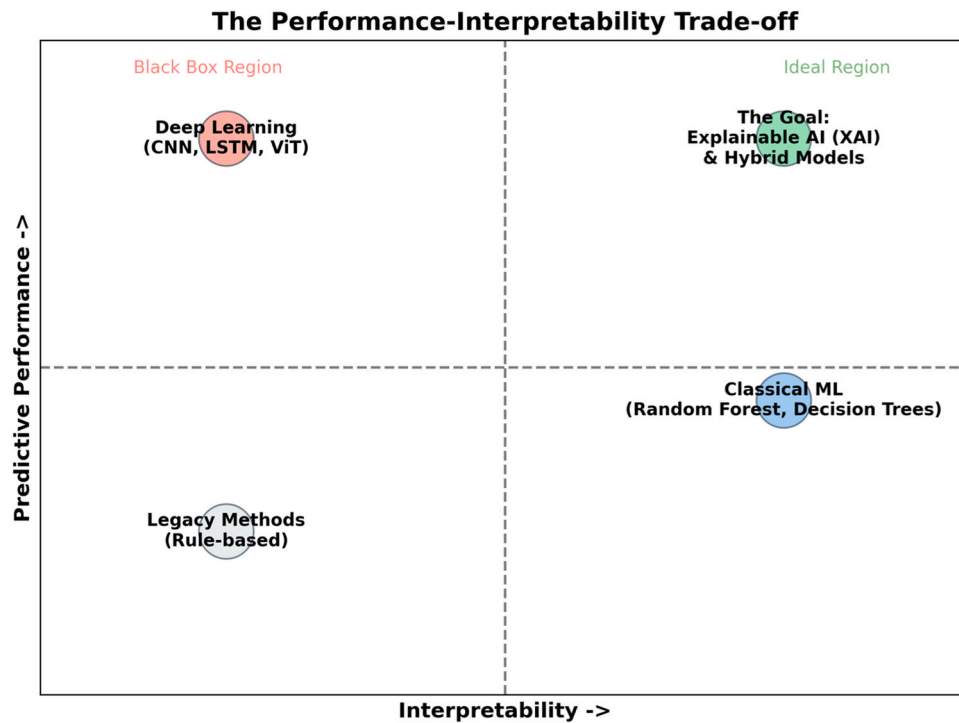


Fig. 7. The trade-off between predictive performance and interpretability across AI methodologies.

when it comes to tasks that are concerned with video analyses related to facial videos as well as multimodal conversation tracking data. Models such as 3D CNNs and transformer models are highly relevant for capturing mental health dynamics, especially when it comes to forecasting conditions like depression or anxiety since mental health dynamics are not static.

For application in limited resource settings, there are ensembles of methods that combine lightweight machine learning models with either PCA or wavelet transformations, making these applicable for mobile-first or wearables-driven stress recognition tasks. These have been

explored with heart-rate variability or EEG patterns, and further work is required for their application as universal, real-time screening tools.^{79,80}

Recent work has further highlighted the benefits of cross-modal transformers, which are capable of handling language, expression, and biosignals data contemporaneously. They demonstrate early promise in terms of accuracy and robustness, especially where one of the data streams can be prone to errors. The future will see early fusion techniques, through the use of attention, incorporated into future architectures.

There exists the possibility of using large language models that have

Late Fusion vs. Hybrid Attention Architectures

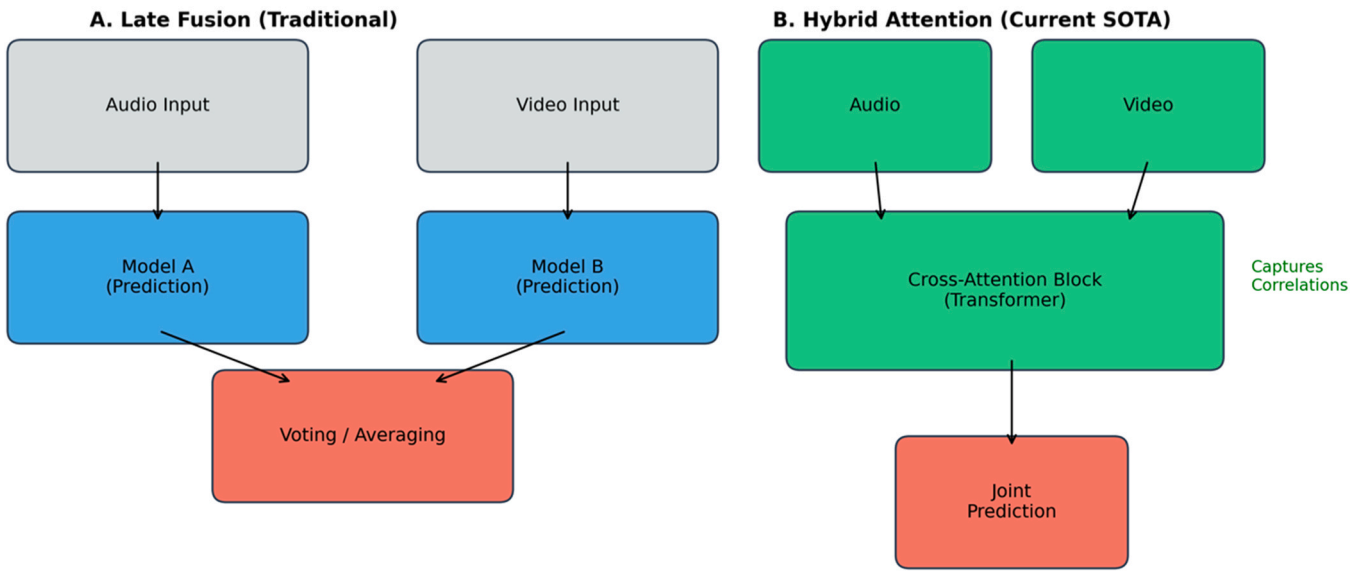


Fig. 8. Comparison of the methodology of traditional and state-of-the-art hybrid architecture.

been fine-tuned from clinical transcripts, mental health forums, or therapy sessions for tasks such as triage, risk prediction, as well as behavior simulation that can be used for the training of mental health professionals. Nonetheless, these goals can be achieved with proper consideration of the accompanying ethics.

Explainability frameworks still lack development and need to move from being more algorithmically grounded to a more reasoned approach. Future models need to incorporate explainability during the learning process by leveraging attention weights, symbolic reasoning, or feedback using saliency maps. The resulting explanations must then be validated by clinical experts and compared to formal standards of care to verify the accuracy of these so-called black box approaches.

Realistic conditions must be given high priority in the modeling of physiological signals. The existing approach and pipeline have proven very successful in a controlled lab environment but tend to perform poorly under the conditions of movement artifacts and inconsistencies in electrodes. Models need to be tested with noisy datasets from wearables for EEG and ECG.

Future systems should seek to not only detect the presence of mental health problems but also to express the severity of these problems as a quantity and consider the issue of comorbidity. Multi-label or multi-task learning algorithms that make predictions about the combination or the scores for the severities of the problems will more realistically capture the complexities of the classification task than traditional binary classification algorithms. On the other hand, it is necessary to take into account the deployment issues of latency, energy consumption, and privacy in the design of the proposed model. Testing the performance of the model in real deployment scenarios like edge computing settings will be imperative in real-world applications.⁸¹ Similarly, integration in the clinical setting will involve testing in actual mental health practice settings.

10. Conclusion

The growing availability of physiological, behavioral, and language data has rapidly supported the increase in the development of ML and DL models for stress and depression detection. This paper has analyzed a wide range of approaches across modalities, including structured self-reports, EEG signals, facial expressions, and social media text. Each

modality brings distinct advantages and challenges, with traditional machine learning models giving the benefits of interpretability and scalability, while deep and multimodal models offer richer representational power and higher performance in complex tasks. The latter methods appear to be the front-runners in the future research in this domain.

Despite the recent advances, many models are still being experimented and tested within a single modality, and only a few incorporate temporal modeling or multimodal fusion. Labeling quality and standardization across datasets are uneven, which impedes reproducibility and benchmarking. The field also lacks consistent approaches to interpretability and clinical validation without which real-world adoption is still far away in the horizon.

Nevertheless, the increasing integration of cross-modal transformers, spatiotemporal architectures, and fine-tuned LLMs signals a shift toward more comprehensive and context-aware systems. There is a clear movement toward models that are not only more accurate but also more transparent, adaptable, and clinically meaningful. To fully realize the promise of computational mental health diagnostics, future research must prioritize the state of the art to ensure accessibility to mental health improvement across a wide range of population.

CRedit authorship contribution statement

Aum Pandya: Writing – original draft. **Nirvaan Singhal:** Writing – review & editing, Methodology. **Prisha Panchal:** Formal analysis, Data curation. **Janmejay Patel:** Visualization, Resources. **Lamya Patel:** Conceptualization. **Vipanshi Thesia:** Writing – review & editing, Investigation. **Merik Patel:** Validation, Supervision.

Authors contribution

All the authors make substantial contribution in this manuscript. AP, NS, PP, JP, LP, VT, and MP participated in drafting the manuscript. AP, NS, PP, JP, LP, VT, and MP wrote the main manuscript, all the authors discussed the results and implication on the manuscript at all stages.

Ethics approval and consent to participate

Not applicable.

Consent for publication

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All relevant data and material are presented in the main paper.

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